

Triangle Half-Rectangle Match

Unit 5: Architecture Firm • Lesson 5-3 • EduWonderLab

UNIT 5 • LESSON 5-3 • TEACHER NOTES

Standard 6.G.1

FORMAT Matching + prove game	TIME 25–30 min
GROUPING Pairs	TOPIC Area of Triangles
STANDARD 6.G.1	PREP LEVEL Light — print triangle and rectangle cards, grid paper

Learning goal *I can find the area of a triangle and explain why it is half of a rectangle.*

Teacher setup

- Print one card set per pair. Students match each triangle with a rectangle of the same base and height.
- They prove the triangle is half, then solve the area cards.
- Include a misconception card about slanted sides.

Materials

Triangle/rectangle cards, Grid paper, Formula reminder, Pencil.

Differentiation & language support

- Formula reminder: $A = \frac{1}{2} \times \text{base} \times \text{height}$ (SPED).
- Sentence stem: “The triangle is half of the rectangle because ____.”
- ESOL: trace the right angle that marks the height.

Key vocabulary (English / Español):

Term	Español	What it means
triangle	triángulo	a three-sided figure
height	altura	the perpendicular distance to the base

Quick teacher check: *Watch for students using the slanted side as height (Card 8).*

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Name: _____ Date: _____ Class: _____

Your goal: *I can find the area of a triangle and explain why it is half of a rectangle.*

What you need

Triangle/rectangle cards, Grid paper.

How to play

1. Match a triangle to a rectangle with the same base and height.
2. Show that the triangle is half the rectangle.
3. Use $A = \frac{1}{2} \times \text{base} \times \text{height}$ on each card.
4. Record the area with square units.

Triangle Cards

Match each triangle to its rectangle, then use $A = \frac{1}{2} \times \text{base} \times \text{height}$.

1. base = 8, height = 5	2. base = 10, height = 6
3. base = 12, height = 4	4. base = 7, height = 6
5. base = 9, height = 8	6. base = 14, height = 5
7. base = 6, height = 11	8. Trap card: base = 12, slanted side = 13, height = 5. Area?

Record your work ($\frac{1}{2} \times \text{base} \times \text{height}$):

Explain your thinking

Explain why the area of a triangle is half the area of a rectangle with the same base and height.

Sentence frame: *The triangle is half of the rectangle because ____.*

★ **Challenge:** Draw two different triangles that both have an area of 24 sq units.

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. base = 8, height = 5 Answer: 20 sq units	2. base = 10, height = 6 Answer: 30 sq units
3. base = 12, height = 4 Answer: 24 sq units	4. base = 7, height = 6 Answer: 21 sq units
5. base = 9, height = 8 Answer: 36 sq units	6. base = 14, height = 5 Answer: 35 sq units
7. base = 6, height = 11 Answer: 33 sq units	8. Trap card: base = 12, slanted side = 13, height = 5. Area? Answer: 30 sq units (use height 5, not 13)

Teaching notes & look-fors

- Card 8 tests the slanted-side error: use the perpendicular height (5).
- Two copies of a triangle form a rectangle (or parallelogram).